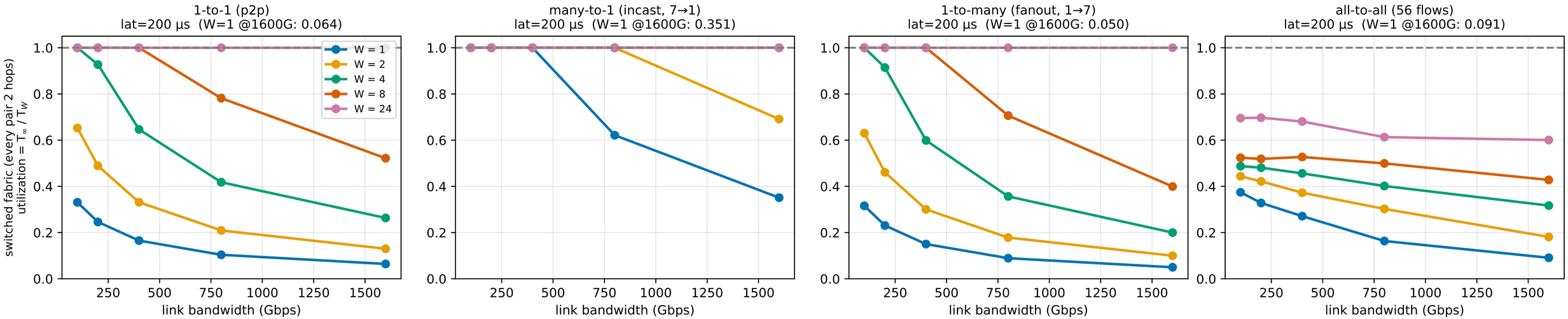
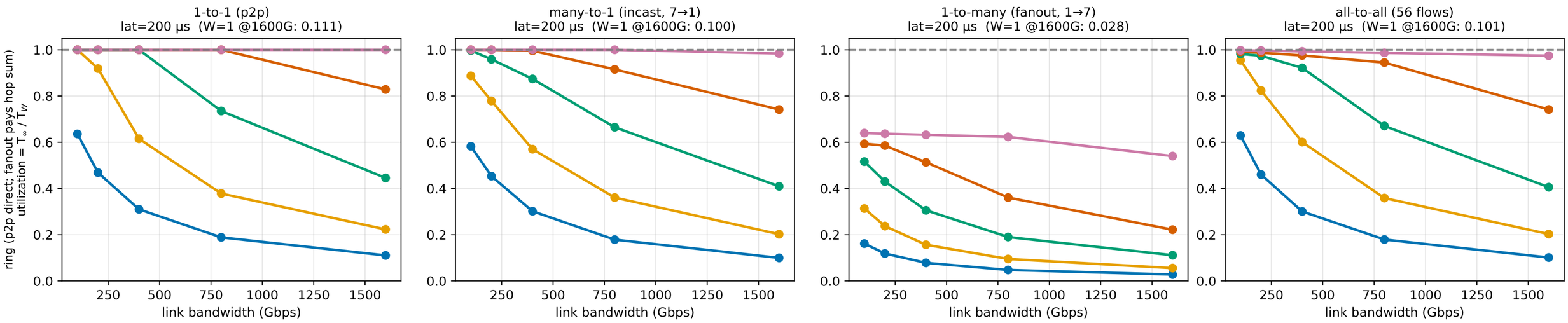
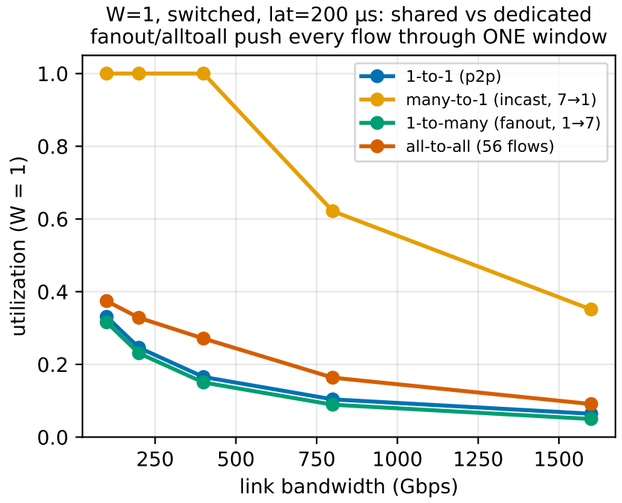
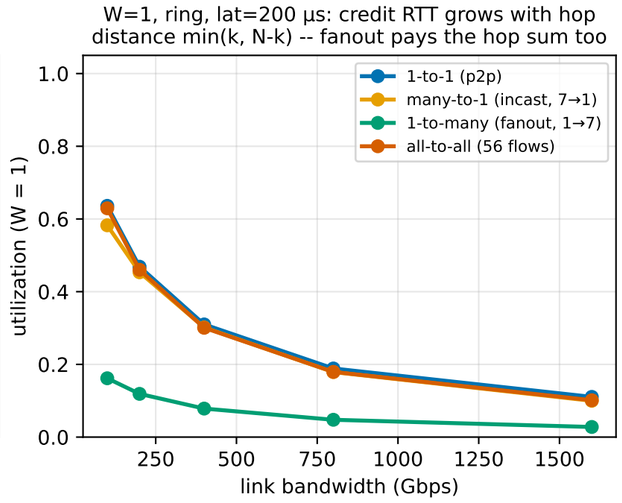
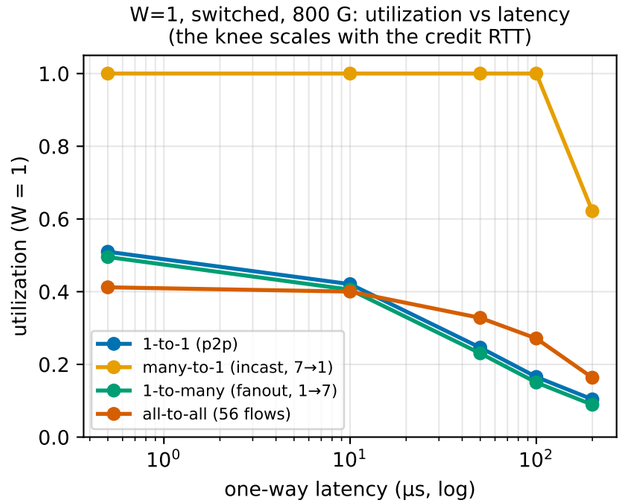
Credit flow control across traffic patterns on the Su fabric (8-GPU, 256 MiB per flow, ring + switched):
window × chunk vs bandwidth-delay product — dedicated (p2p/incast) vs shared (fanout/alltoall) credit windows











Raw times @800 G, lat=200 µs, W=1 (ms)

switched	1-to-1 (p2p)	30.58	(1.00x p2p)
switched	many-to-1 (incast, 7→1)	31.09	(1.02x p2p)
switched	1-to-many (fanout, 1→7)	216.47	(7.08x p2p)
switched	all-to-all (56 flows)	217.12	(7.10x p2p)
ring	1-to-1 (p2p)	15.28	
ring	many-to-1 (incast, 7→1)	61.14	
ring	1-to-many (fanout, 1→7)	247.55	
ring	all-to-all (56 flows)	247.61	

BDP law per FLOW: W·chunk/(ser + 2·lat).
One VC per endpoint -> fanout and alltoall
share ONE window across their flows, so the
sender serializes 224 chunks through W:
W=1 switched = 7.1x p2p. On the ring the
credit RTT to rank_k costs min(k, N-k) hops,
so fanout also pays the hop sum (224 chunks
x hop-distance sum 16 = 247.7 ms, matched).